

Exercise 2.24. Prove that $m \equiv n \pmod{15}$ if and only if $m \equiv n \pmod{3}$ and $m \equiv n \pmod{5}$. To do this, here are the two things that you should prove:

(a) If $m \equiv n \pmod{15}$, then $m \equiv n \pmod{3}$ and $m \equiv n \pmod{5}$.

(b) If $m \equiv n \pmod{3}$ and $m \equiv n \pmod{5}$, then $m \equiv n \pmod{15}$.

Solution.

Let us prove both directions.

Recall that

$$m \equiv n \pmod{k}$$

means that

$$k \mid (m - n).$$

We want to prove:

$$m \equiv n \pmod{15} \iff m \equiv n \pmod{3} \text{ and } m \equiv n \pmod{5}.$$

(a) If $m \equiv n \pmod{15}$, then $m \equiv n \pmod{3}$ and $m \equiv n \pmod{5}$

Suppose

$$m \equiv n \pmod{15}.$$

Then

$$15 \mid (m - n).$$

Since $3 \mid 15$ and $5 \mid 15$, it follows that

$$3 \mid (m - n) \quad \text{and} \quad 5 \mid (m - n).$$

Therefore,

$$\boxed{m \equiv n \pmod{3}}$$

and

$$\boxed{m \equiv n \pmod{5}}.$$

(b) If $m \equiv n \pmod{3}$ and $m \equiv n \pmod{5}$, then $m \equiv n \pmod{15}$

Suppose

$$m \equiv n \pmod{3} \quad \text{and} \quad m \equiv n \pmod{5}.$$

Then

$$3 \mid (m - n) \quad \text{and} \quad 5 \mid (m - n).$$

Because 3 and 5 are distinct primes, Exercise 2.23 tells us that if both 3 and 5 divide an integer, then their product divides that integer. Hence,

$$3 \cdot 5 = 15 \mid (m - n).$$

Therefore,

$$\boxed{m \equiv n \pmod{15}}.$$

Thus,

$$\boxed{m \equiv n \pmod{15} \iff m \equiv n \pmod{3} \text{ and } m \equiv n \pmod{5}}.$$